

A-MC	B-FAU	B-ENG	B-NOY	B-HAY	C-CAR	TOTAL
/20	/19	/10	/20	/16	/15	/100

Student Name

Sydney Technical High
School

Teacher _____

2012

TRIAL HIGHER SCHOOL
CERTIFICATE
EXAMINATION

Chemistry



General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Approved calculators may be used
- Write your student number in the space provided

Total marks – 100

Section I Pages 2 - 28

85 marks

This section has two parts, Part A and Part B

Part A – 20 marks

- Attempt Questions 1-20
- Allow about 35 minutes for this part

Part B – 65 marks

- Attempt Questions 21-34
- Allow about 1 hour and 40 minutes for this part

Section II Pages 29 -30

15 marks

- Attempt Questions 35
- Allow about 45 minutes for this section

Student Number _____

Section I

85 marks

Part A – 20 marks

Attempt Questions 1-20

Allow about 35 minutes for this part

Use the multiple-choice answer sheet.

Select the alternative A,B,C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

A  B  C  D 

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A  B  C  D 

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A  B  C  D 

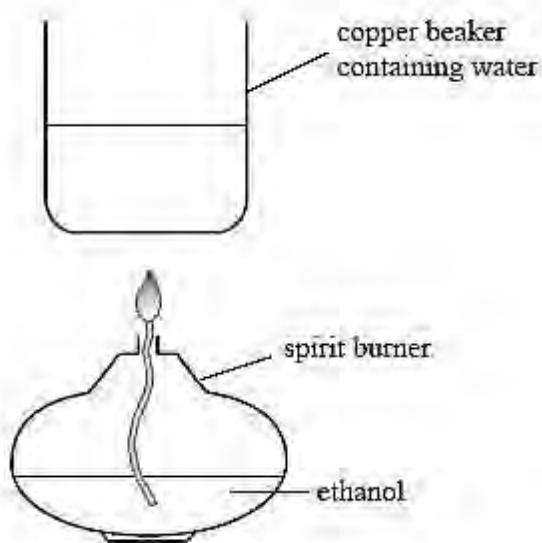
correct

Multiple Choice Answer Sheet

1.	A ☐	B ☐	C ☐	D ☐
2.	A ☐	B ☐	C ☐	D ☐
3.	A ☐	B ☐	C ☐	D ☐
4.	A ☐	B ☐	C ☐	D ☐
5.	A ☐	B ☐	C ☐	D ☐
6.	A ☐	B ☐	C ☐	D ☐
7.	A ☐	B ☐	C ☐	D ☐
8.	A ☐	B ☐	C ☐	D ☐
9.	A ☐	B ☐	C ☐	D ☐
10.	A ☐	B ☐	C ☐	D ☐
11.	A ☐	B ☐	C ☐	D ☐
12.	A ☐	B ☐	C ☐	D ☐
13.	A ☐	B ☐	C ☐	D ☐
14.	A ☐	B ☐	C ☐	D ☐
15.	A ☐	B ☐	C ☐	D ☐
16.	A ☐	B ☐	C ☐	D ☐
17.	A ☐	B ☐	C ☐	D ☐
18.	A ☐	B ☐	C ☐	D ☐
19.	A ☐	B ☐	C ☐	D ☐
20.	A ☐	B ☐	C ☐	D ☐

1. Which of the following pairs of reactants could undergo an addition reaction to produce 2-bromobutane?
- (A) bromine and 2-butene
 - (B) bromine and butane
 - (C) hydrogen bromide and 2-butene
 - (D) hydrogen bromide and butane
2. What is the molecular formula of the *monomer* used in the production of polystyrene?
- (A) C_2H_4
 - (B) $\text{C}_2\text{H}_3\text{Cl}$
 - (C) $\text{C}_6\text{H}_{12}\text{O}_6$
 - (D) C_8H_8
3. Four chemical reactions are shown below:
- I $\text{MgO} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2\text{O}$
 - II $\text{Cu} + 2\text{AgNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + 2\text{Ag}$
 - III $\text{CuCO}_3 \rightarrow \text{CuO} + \text{CO}_2$
 - IV $4\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$
- Which of the reactions are redox reactions?
- (A) II only
 - (B) II and IV
 - (C) I and II
 - (D) III and IV
4. What is the maximum mass of ethanol that can be produced by the fermentation of 0.75 kg of glucose?
- (A) 192 g
 - (B) 384 g
 - (C) 575 g
 - (D) 750 g

5. A student set up the following apparatus to compare the heat of combustion of methanol, ethanol and 1-propanol.

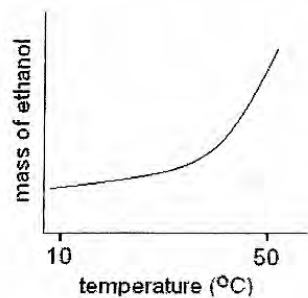


Which one of the following variables, if not controlled carefully, would have the greatest effect on the validity of the procedure?

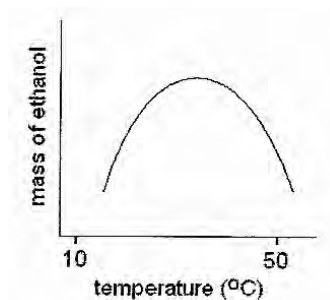
- (A) The distance between the flame and the can.
 - (B) The temperature to which the water is heated.
 - (C) The volume of water heated.
 - (D) The initial temperature of the water.
6. Which of the following lists contains only devices that can be used to detect radiation?
- (A) Smoke detector, photographic film, Geiger counter, particle accelerator
 - (B) Thickness gauge, cloud chamber, photographic film, smoke detector
 - (C) Thickness gauge, scintillation counter, Geiger counter, particle accelerator
 - (D) Cloud chamber, scintillation counter, photographic film, Geiger counter

7. Which of the following graphs represents the mass of ethanol produced when a constant mass of glucose is fermented, for a fixed time interval at different temperatures?

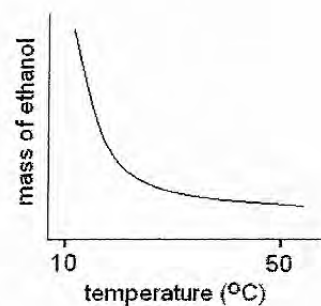
(A)



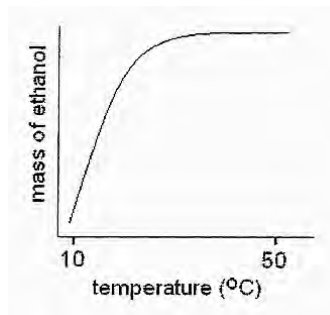
(B)



(C)



(D)



8. This question refers to the following substances:

- i. H_2CO_3
- ii. NH_4Cl
- iii. NaCH_3COO
- iv. CH_3COOH

Solutions of which of the substances listed would turn blue litmus red?

- (A) (i) only
- (B) (i) and (iv)
- (C) (i), (ii) and (iv)
- (D) (i), (iii) and (iv)

9. The following acid-base indicators change colour depending on pH as shown in the following table.

pH \ Indicator	2	4	6	8	10	12	14
Methyl orange	red	Yellow					
Bromothymol blue	Yellow			Blue			

What colour will the following indicators be if a few drops are added to 0.1 M hydrochloric acid, and to 0.1 M acetic (ethanoic) acid (CH_3COOH)?

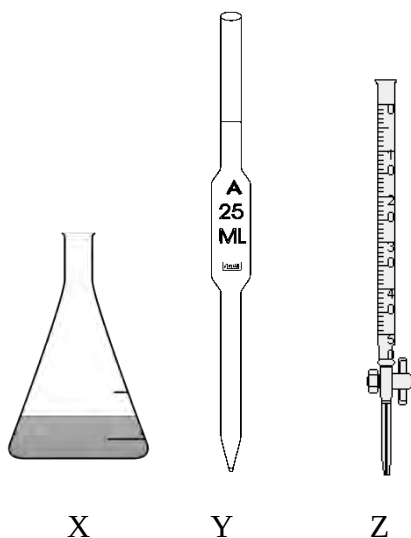
	Methyl orange		Bromothymol blue	
	HCl	CH_3COOH	HCl	CH_3COOH
(A)	Red	Red	Yellow	Yellow
(B)	Red	Yellow	Yellow	Yellow
(C)	Yellow	Yellow	Blue	Blue
(D)	Red	Yellow	Blue	Blue

10. A solution of a strong acid has a pH of 3.2. A student dilutes 10 mL of the solution to 1000 mL. What is the final pH?
- (A) 3.2
(B) 4.2
(C) 5.2
(D) 6.2
11. Solution X contains a monoprotic acid and its pH is 1.50. Solution Y contains a monoprotic acid and its pH is 4.5. Equal volumes of Solution X and Solution Y each neutralise 25.00 mL of 0.100 M HCl.

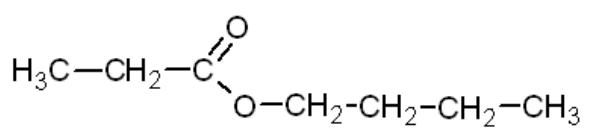
It can be correctly concluded that

- (A) Solution Y contains a weak acid. 4.2
(B) Solution X contains a strong acid.
(C) Solution Y is three times as acidic as Solution X.
(D) Solution X is less concentrated than Solution Y.

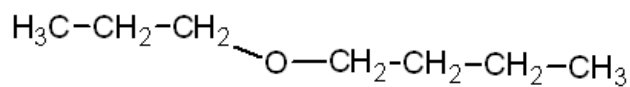
12. Which of the following pieces of equipment may have water left in it before it is used?



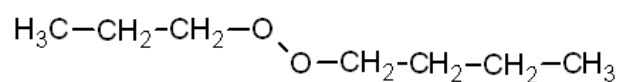
- (A) Y only
(B) X and Y
(C) X only
(D) Y and Z
13. A student mixes 1-butanol with propanoic acid in an esterification reaction.
- Which of the following shows the structure of the organic product?



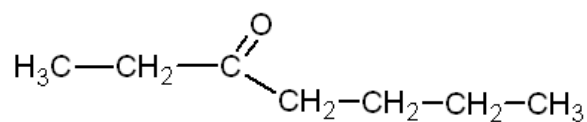
(A)



(B)



(C)



(D)

14. Which combination of solutes forms a buffer solution in water?

- (A) Nitric acid and potassium nitrate
- (B) Citric acid and potassium citrate
- (C) Hydrochloric acid and sodium hydroxide
- (D) Ammonia and potassium nitrate

15. Which of the following is **not** a use of ammonia?

- (A) manufacture of explosives
- (B) manufacture of detergents
- (C) manufacture of fertilisers
- (D) manufacture of paper

16. A student wants to determine the sulfate content of a fertilizer.

Following a typical procedure, they obtain the results shown below.

Mass of fertilizer used (g)	2.34
Mass of fertilizer that did not dissolve (g)	0.18
Volume of saturated $\text{BaCl}_2(\text{aq})$ added (mL)	50
Mass of glass filter (g)	19.5
Mass of glass filter with dry BaSO_4 (g)	21.6

What is the sulfate content of the fertilizer?

- (A) 40%
- (B) 50%
- (C) 75%
- (D) 90%

17. The molecules O_2 and O_3 are:

- (A) diatomic
- (B) compounds
- (C) isotopes
- (D) allotropes

18. Identify the main source of the pollutant *carbon monoxide*, in the lower atmosphere.

- (A) Burning of fossils fuels
- (B) Anaerobic decomposition of organic matter
- (C) Smelting of metal ores
- (D) Deforestation

19. A solid sample was known to contain two calcium salts. In order to determine the anions present, some tests were done on the solid, producing the following results.

Test done	Results obtained
Observation of colour	White
Addition of water to solid	Solid partially dissolved
Addition of barium chloride to solution	No precipitate
Addition of silver nitrate to solution	White precipitate
Addition of HCl(aq) to solid	Gas bubbles observed

Which two ions were present in the sample?

- (A) PO_4^{3-} and SO_4^{2-}
- (B) Cl^- and SO_4^{2-}
- (C) CO_3^{2-} and Cl^-
- (D) CO_3^{2-} and PO_4^{2-}

20. In which layer of the earth's atmosphere do oxides of sulfur occur naturally?

- (A) troposphere
- (B) stratosphere
- (C) mesosphere
- (D) ionosphere

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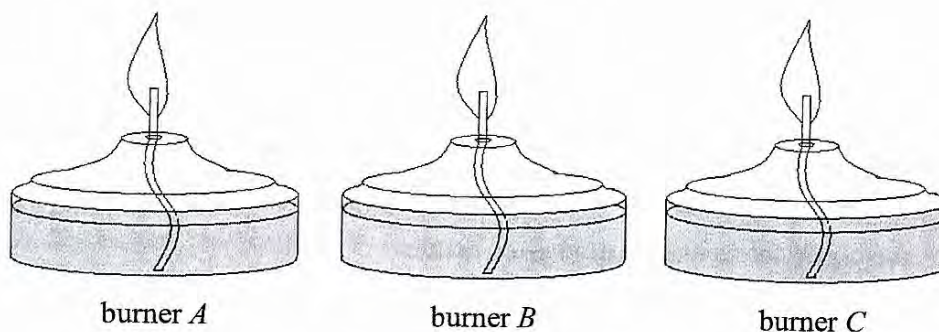
Section I (continued)**Part B – 65 marks****Attempt Questions 21 - 29****Allow about 1 hour and 40 minutes for this part**

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 21 (6 marks)**Marks**

Three different fuels, butan-1-ol, ethanol and octan-1-ol, were burned in three separate identical spirit burners.



The fuel burned in burner B produced a bright yellow, very smoky flame, while the fuels in burners A and C produced pale blue flames with little smoke.

- (a) Identify the fuel burned in burner B and account for the smoky appearance of the flame, in terms of the combustion products.

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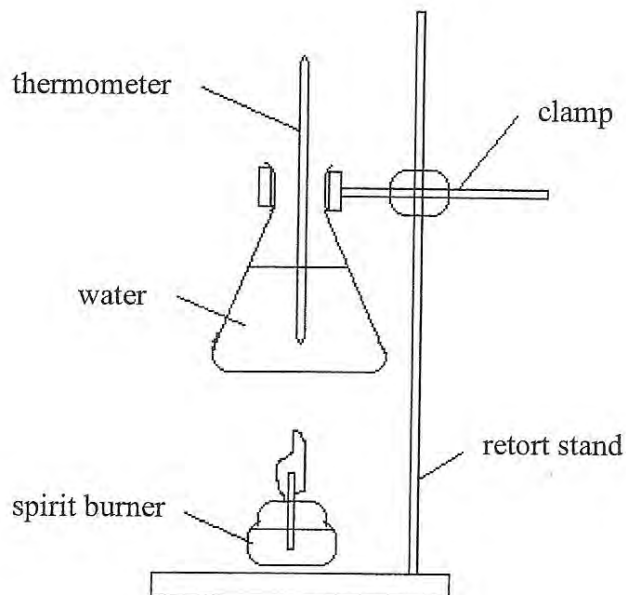
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Question 21 continues on page 14.

Question 21 (continued)

Marks

- (b) Burners A and C were each used to heat water in identical experimental apparatus, as shown below.



Measurements recorded during the experiment are shown in the following table.

Burner	Mass of water heated (g)	Initial temperature of water ($^{\circ}\text{C}$)	Final temperature of water ($^{\circ}\text{C}$)	Initial mass of burner (g)	Final mass of burner (g)
A	200	20.0	30.0	275.48	274.83
B	200	20.0	30.0	287.61	286.51

Compare the heat of combustion (per gram) for these two fuels and use this information to identify the fuel in burner A

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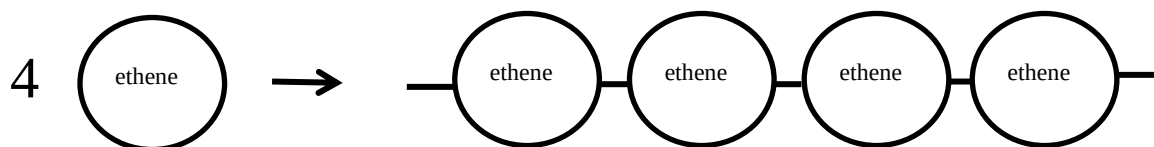
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Question 22 (3 marks)**Marks**

The following model was used by a chemistry student to demonstrate their understanding of the reaction to form a short segment of polyethene, and the composition of polyethene.



With reference to the model only, assess the student's understanding of polyethene and the reaction that forms it.

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Question 23 (2 marks)

Radioisotopes are used in medicine.

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Describe the way a named radioisotope is used in medicine.

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Question 24 (4 marks)

Discuss the production and use of a named biopolymer (other than cellulose) and the need for the further development of biopolymers.

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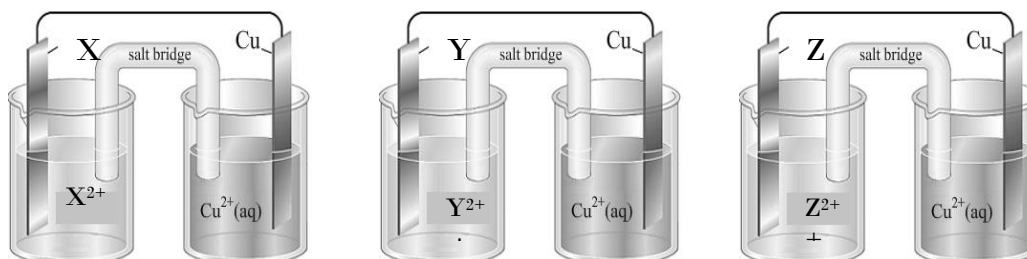
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Question 25 (4 marks)

Half-cells made from three metals (X, Y and Z) and their solutions were coupled with a copper half-cell under standard conditions, as shown in the diagram below.

Marks

The voltage produced and the polarity of the copper electrode, were recorded in a table below.

Metal	Cell voltage produced by coupling metal half- cell with Cu half-cell	Polarity of Copper
X	0.31	+ve
Y	1.14	-ve
Z	0.42	-ve

- (a) Outline the function of the salt bridge in the cells above.

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- (b) Using the data provided, rank the metals (X, Y, Z and Cu), in increasing order of reactivity.
Explain how you arrived at your answer.

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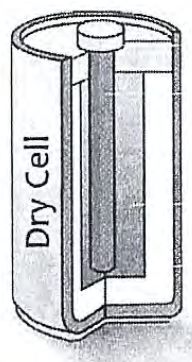
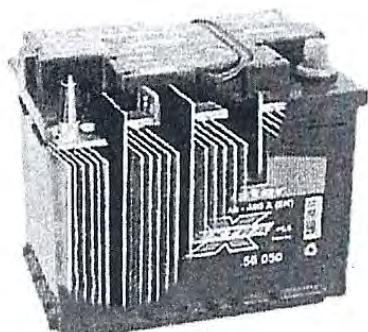
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ENGEL /10

Question 26 (4 marks)**Marks**
4

The images below show two types of common battery cells.



Describe the structure and chemistry of *either* cell.

Include chemical equations in your answer.

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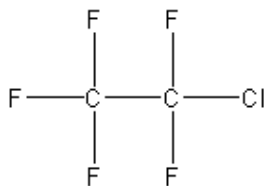
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Marks**Question 27 (6 marks)**

Relatively recent human activity has reduced the concentration of ozone in the upper atmosphere, with serious consequences.

(a) The molecules responsible for ozone depletion are CFCs.

(i) Give the systematic name of the compound below.

1

(ii) Describe how a compound such as that shown in part (i) can destroy ozone. Support your answer with chemical equations.

3

(b) Outline the consequences of ozone depletion in the upper atmosphere.

2

NOYES

/ 20

Question 28 (4 marks)

A can of soda water contains carbon dioxide dissolved under pressure, forming carbonic acid.

Marks

- (a) With the aid of a chemical equation, explain why carbonic acid is described as a weak acid.

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- (b) Using Le Chatelier's Principle, justify the change in pH of the contents, after a can of soda water is opened.

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Question 29 (10 marks)

A student used temperature change during neutralization to calculate the concentration of hydrochloric acid. The method they used was:

Marks

1. 1L of 0.145 M NaOH and 1L of HCl were allowed to sit at room temperature for 60 minutes.
2. 25.0 mL of 0.145 M NaOH was added to a polystyrene cup using a volumetric pipette.
3. The temperature of the NaOH(aq) was measured using a thermometer.
4. 10.0 mL of HCl was added to the cup using a volumetric pipette.
5. The highest temperature reached was measured.
6. Steps 1 – 4 were repeated with 20, 30, 40, 50 and 60 mL of HCl.

The results they obtained are shown in the table below.

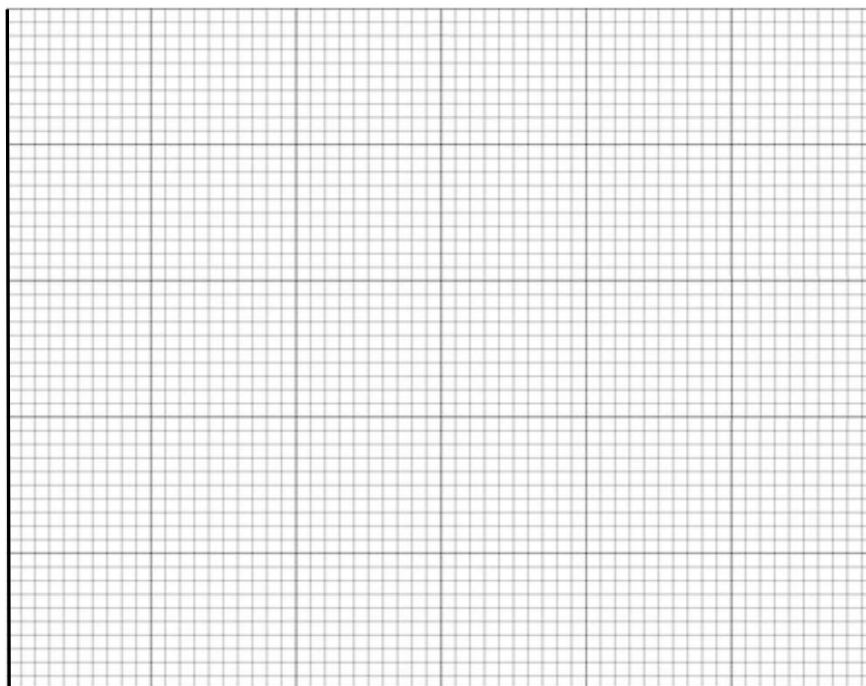
V (HCl) added vs temp. when neutralizing 0.145 M NaOH.

Volume of HCl added (mL)	Max. temperature reached ($^{\circ}\text{C}$)
0	21.0
10	28.0
20	35.0
30	35.0
40	32.5
50	30.0
60	27.5

2

- (a) Use the grid below to graph the data above.

Temperature ($^{\circ}\text{C}$)



3

0

Volume HCl added (mL)

- (b) Draw two straight lines through the points and extend them until they cross.

1

Question 29 continues on page 23.

Question 29 (continued)

- (c) What volume of HCl was required to completely neutralize the NaOH? **Marks**
On your graph, show how you obtained this value.

..... **1**

- (d) Calculate the concentration of the HCl.

..... **2**

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- (e) The NaOH was standardised using a 0.1 M oxalic acid primary standard.

Outline how the 0.1 M oxalic acid ($\text{H}_2\text{C}_2\text{O}_4$) standard solution was prepared.

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Question 30 (6 marks)

An ester can be prepared by using 1-propanol and pentanoic acid.

**Mark
s**

(a) Identify a use for esters.

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(b) Write a structural equation for the formation of this ester.

2

(c) Outline with the aid of a diagram how this ester could be prepared in the school laboratory.

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HAYES	/16
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Question 31 (4 marks)

Hydrogen sulfide gas is extremely flammable, has an unpleasant smell similar to rotten eggs and is toxic.

Mark

Hydrogen sulfide gas is formed when hydrochloric acid reacts with sodium sulfide. In an experiment, 0.15 g of sodium sulfide was added to 25.0 mL of 0.10 M HCl.

Calculate the maximum volume of hydrogen sulfide gas that could be produced by this reaction at 25° C and 100 kPa?

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4**Question 32** (1 mark)

Identify the bonding within ozone, using a Lewis electron-dot diagram.

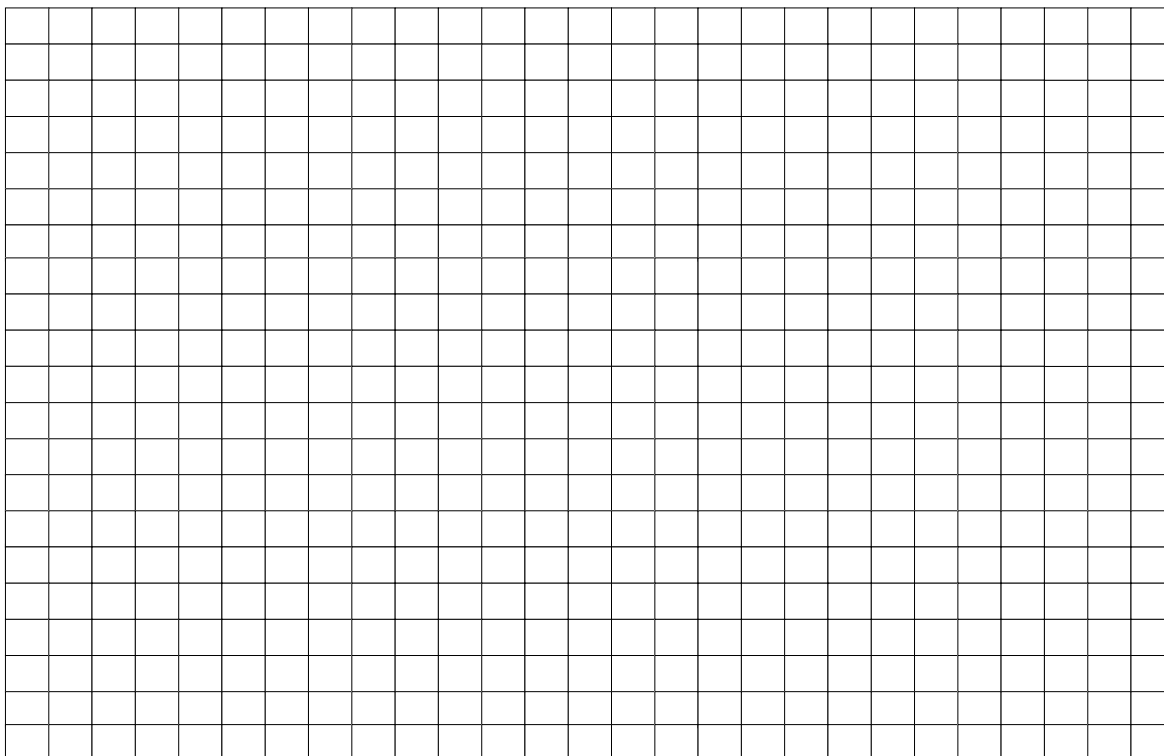
1

Question 33 (6 marks)**Mark**

A student gathered the following data about the Haber process, from secondary sources.

Pressure (MPa)	0	10	20	30	40	50
% Yield of ammonia at 400°C	0	17	34	48	64	79
% Yield of ammonia at 550°C	0	4	10	14	19	24

(a) Graph *the yield* (vs. pressure) for each temperature.

3

(b) By referring to the graphs, and including a reaction equation, explain the effects of temperature and pressure on the yield of ammonia.

3

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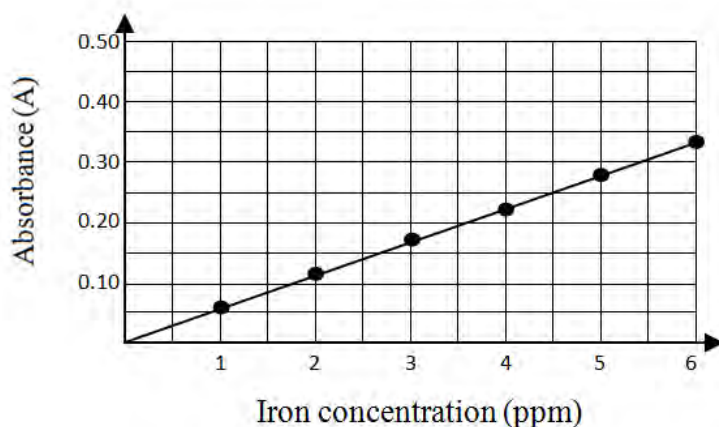
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Question 34. (5 marks)

Atomic absorption spectroscopy (AAS) is widely used by analytical chemists to detect and measure the concentration of metal ions in solution. The level of iron in a sample of mineral water was analysed using AAS. The calibration graph shown below was used to determine the concentration of iron in the sample.

Mark

- (i) The absorbance (A) of the mineral water sample was 0.25
Use the graph determine the concentration of iron in the mineral water.

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1

- (ii) Calculate the mass of iron that would be present in 600mL of the mineral water.

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2

- (iii) Outline how the chemist would have obtained the calibration graph before testing the sample of mineral water.

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2

Section II

Total marks: 15

Attempt \Questions 35

Allow about 45 minutes for this Section.

Answer the OPTION question on the writing paper supplied. Extra writing paper is available.

Show all relevant working in questions involving calculations.

	Page
Question 35 – Industrial Chemistry	30

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Question 35 - Industrial Chemistry (15 marks)**Marks**

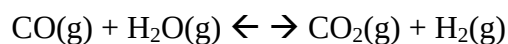
- (a) Many industrial processes involve equilibrium reactions.

Manipulating equilibrium conditions is an important part of industrial chemistry.

1

- (i) Identify the experimental variable that changes the value of the equilibrium constant.

- (ii) The water-gas shift reaction is a useful industrial method for the production of hydrogen gas, some of which is used to provide hydrogen for the Haber process. The equation for the process is:



A flask initially containing only 1.3 moles/litre of CO and 2.4 moles/litre of H₂O was found to contain 0.6 moles/litre of CO₂ at equilibrium.

2

Calculate the equilibrium constant K for the reaction.

- (b) Environmental considerations have become increasingly important in chemical industry. The production of sulfuric acid has environmental impacts.

1

Outline an environmental issue associated with the extraction of sulfur.

- (c) Describe and explain the exothermic nature of sulfuric acid ionization.

2

- (d)

- (i) Models are often used to explain complex concepts. Outline a first-hand investigation that you undertook to model an equilibrium reaction.

2

- (ii) Assess the validity of the information that could be collected in this investigation.

2

- (e) Discuss the issues associated with shrinking world resources with regard to one identified natural product that is not a fossil fuel.

5

Identify the replacement materials used and/or current research in place to find a replacement for the named material.

PART A: Multiple Choice

1. C

2. D

3. B

4. B

5. A

6. D

7. B

8. C

9. B

10.

11. A

12. C

13. A

14. B

15. D

16.

17. D

18. A

19. C

20. A

PART B: FAULDER (19 marks)

Question 21 (6 marks)

(a)

<u>Marking Criteria</u>	<u>Marks</u>
- Identifies the fuel burned in burner B - Accounts for smokey flame – incomplete combustion - Carbon produced	3
- Identifies the fuel burned in burner B - Names either carbon or incomplete combustion as cause of smokey flame	2
Identifies either the fuel burned in burner B OR names carbon OR incomplete combustion as cause of smokey flame	1

Sample answer

Burner B contained octan-1-ol

Incomplete combustion results in the production of solid carbon (soot) which causes a smokey flame.

(b)

<u>Marking Criteria</u>	<u>Marks</u>
- Calculates the heats of combustion of the alkanols - Identifies A as butan-1-ol	3
Calculates the heats of combustion of the alkanols only OR Calculates one heat of combustion and correctly identifies A	2
Calculates the heat of combustion of one alkanol correctly OR Correctly identifies A	1

$$\begin{aligned} \text{A} \quad & 200 \times 4.18 \times 10 / 0.65 \\ & = 12.86 \text{kJ/g} \end{aligned}$$

$$\begin{aligned} \text{C} \quad & 200 \times 4.18 \times 10 / 1.1 \\ & = 7.6 \text{kJ/g} \end{aligned}$$

Therefore A contains butan-1-ol (the greater the chain length the greater the energy released)

Question 22 (3 marks)

<u>Marking Criteria</u>	<u>Marks</u>
- Assesses (judgement) the student's understanding - refers to the diagram to describe a feature of the model - describes the correct features of addition polymerisation and the composition of polyethene.	3
- Describes the correct features of addition polymerisation and the composition of polyethene AND gives a judgement OR - Describes a feature of polyethene shown by the model AND gives a judgement OR - Describes the correct features of addition polymerization and the composition of polyethene AND a feature shown by the model	2
Identifies a feature of polyethene the model shows OR a feature/s it doesn't show	1

Sample answer

The model demonstrates the student's understanding which is that polyethene is a long-chained molecule formed from the joining of several smaller molecules of ethene.

However, the student's model does not show that the polymerisation is an addition reaction, where the monomers join as the double bonds in ethene break, allowing the carbon atoms to bond to each other. Thus, the repeating unit is not ethene.

Therefore, overall, the model shows a limited understanding of the composition and reaction to produce polyethene.

Question 23 (2 marks)

<u>Marking Criteria</u>	<u>Marks</u>
Describes the use of a named radioisotope in medicine	2
Identifies a radioisotope used in medicine OR identifies a use of a radioisotope in medicine	1

Sample Answer

Technetium -99m can be used as a liver-bile tracer to determine the functioning of the bile duct.

Question 24 (4 marks)

<u>Marking Criteria</u>	<u>Marks</u>
- Names biopolymer - Discusses production - Discusses use - Discusses need for further development	4
- Names biopolymer AND - Discusses two of production, use and development	3
- Names biopolymer AND - Discusses one only of either production, use or development	2
Names biopolymer Or discusses one of either production, use or development	1

Sample Answer

Polyhydroxybutyrate (PHB) is a biopolymer produced by micro-organisms (eg *Alcaligenes eutrophus*) during fermentation of renewable carbohydrate food stocks.

It has the potential to replace polypropylene for use in packaging such as bottles, bags and wrapping film.

As biopolymers are made from renewable resources and are biodegradable/ biocompatible, further development is needed to reduce production costs and improve properties. This would make the use of biopolymers more viable compared to conventional polymers which are made from non-renewable petroleum resources.

Question 25 (4 marks)

(a)

<u>Marking Criteria</u>	<u>Marks</u>
Outlines the function of the salt bridge in a galvanic cell.	1

Sample Answer

The salt bridge allows ions to move between half-cells to complete the circuit

(b)

<u>Marking Guidelines</u>	<u>Marks</u>
• Determines the correct order of reactivity of the metals and justifies the order fully.	3
• Determines the correct order of reactivity of the metals AND gives one correct relevant feature of the galvanic cells shown OR • two correct relevant feature of the galvanic cells shown.	2
• Determines the correct order of reactivity of the metals OR • a correct relevant feature of galvanic cells.	1

Sample Answer

Galvanic cells are based on redox reactions. The more reactive metal is oxidised and this occurs at the anode which is the negative terminal. The voltage produced by different combinations is an indication of the reactivity, the greater the voltage the greater difference. Metal X – Copper is the positive electrode so metal X is oxidised and hence more reactive than copper

Metal Y and Z – Copper is the negative electrode so it is more reactive than both Y and Z

The cell voltage of Cu and Y is larger than Cu and Z so the difference in reactivity is greater between Cu and Y than Cu and Z. Z must therefore be more reactive than Y.

Rank : Y, Z, Cu, X

PART B: ENGEL (10 marks)

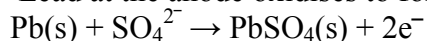
Question 26 (4 marks)

Marking Criteria	Marks
• Describes the anode and the chemical reaction including the correct half-equation for the reaction. • Describes the cathode and the chemical reaction including the correct half-equation for the reaction. • Identifies the electrolyte • States voltage produced • Describes cell rechargeability	4
• Describes the anode/cathode and the chemical reaction AND • Describes the other electrode and the chemical reaction OR • Writes a correct half-equation OR • Identifies the electrolyte OR • States voltage produced OR • Describes cell rechargeability OR	2-3
• Identifies the electrolyte OR • Identifies the cathode or anode OR • Writes a correct half-equation. OR • States voltage produced OR rechargability	1

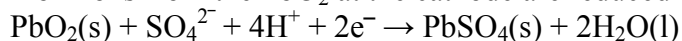
Sample answers

A lead-acid cell is made of a lead anode plate and lead(IV) oxide coated lead cathode plate immersed in a sulphuric acid (35%) electrolyte solution.

Lead at the anode oxidises to form Pb^{2+} ions in the form of solid PbSO_4 .



Pb^{4+} ions from the PbO_2 at the cathode are reduced to Pb^{2+} ions and also form solid PbSO_4 .

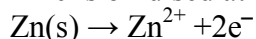


The cell is rechargeable and produces approx. 2 volts.

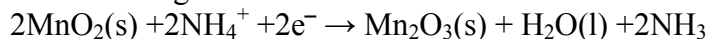
OR

A dry cell has a central positive cathode consisting of an inert graphite rod surrounded by graphite and manganese dioxide powder. The zinc casing for the cell acts as the negative anode. Between the two electrodes is an aqueous electrolyte paste of ammonium chloride 26%(w/w) containing more powdered graphite and manganese dioxide.

Zinc is oxidised at the anode to zinc ions.



Mn^{4+} in manganese dioxide is reduced to Mn^{3+} when Mn_2O_3 solid is produced at the cathode.



It is non-rechargeable and produces 1.5 volts

Question 27 (6 marks)

27(a)(i)

Marking Criteria	Marks
Correctly names the compound.	1

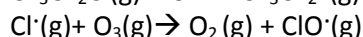
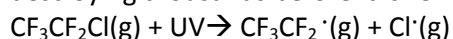
1-chloro-1,1,2,2,2-pentafluoroethane.

27(a)(ii)

Marking Criteria	Marks
Outlines how the compound can destroy ozone, including relevant chemical equations	3
- Outlines how the compound in a(i) can destroy ozone, OR - Includes relevant chemical equations, OR - Incomplete outline and one relevant equation	2
One correct statement or relevant equation.	1

Sample answer

The compound shown can destroy ozone because in the presence of UV light the C-Cl bond is broken, producing a chlorine free radical. This Cl radical attacks ozone, and in a sequence of reactions shown below, is regenerated. It is able to continue reacting with other ozone molecules, destroying thousands before it is removed from the atmosphere.



$\text{ClO}\cdot(\text{g}) + \text{O}\cdot(\text{g}) \rightarrow \text{Cl}\cdot(\text{g}) + \text{O}_2(\text{g})$, and thus the Cl radical is regenerated and continues to destroy O_3 molecules.

27(b)

Marking Criteria	Marks
Identifies ozone depletion results in increased UV radiation reaching the Earth, AND one consequence of this increased UV radiation	2
- Identifies that ozone depletion results in increased UV radiation reaching the Earth, OR - Identifies one consequence of increased UV radiation reaching the earth.	1

Sample answer

Ozone depletion results in increased UV radiation reaching the Earth. The consequences of this include: increased rates of skin cancers and eye cataracts, phytoplankton damage or destruction resulting in disruption of ecosystems.

PART B: NOYES (20 marks)

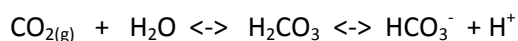
Question 28 (4 marks)

28.a.

Marking Guidelines	Marks
- Explains a weak acid - Equation shows partial ionization	2
- Describes a weak acid as partial ionisation OR - Equation does not show partial ionization	1

Sample answer:

Carbonic acid is only slightly ionised in water, producing a low concentration of hydrogen ions.



28.b.

Marking Guidelines	Marks
- States Le Chatelier's Principle - Predicts the change in equilibrium - Describes the change in H^+ and pH	2
- States Le Chatelier's Principle OR - Predicts the change in equilibrium OR - Describes the change in H^+ and pH	1

Sample answer:

Le Chatelier's Principle states that when a system reaches equilibrium and a change occurs in one of the conditions then the system adjusts so that it can return to equilibrium.

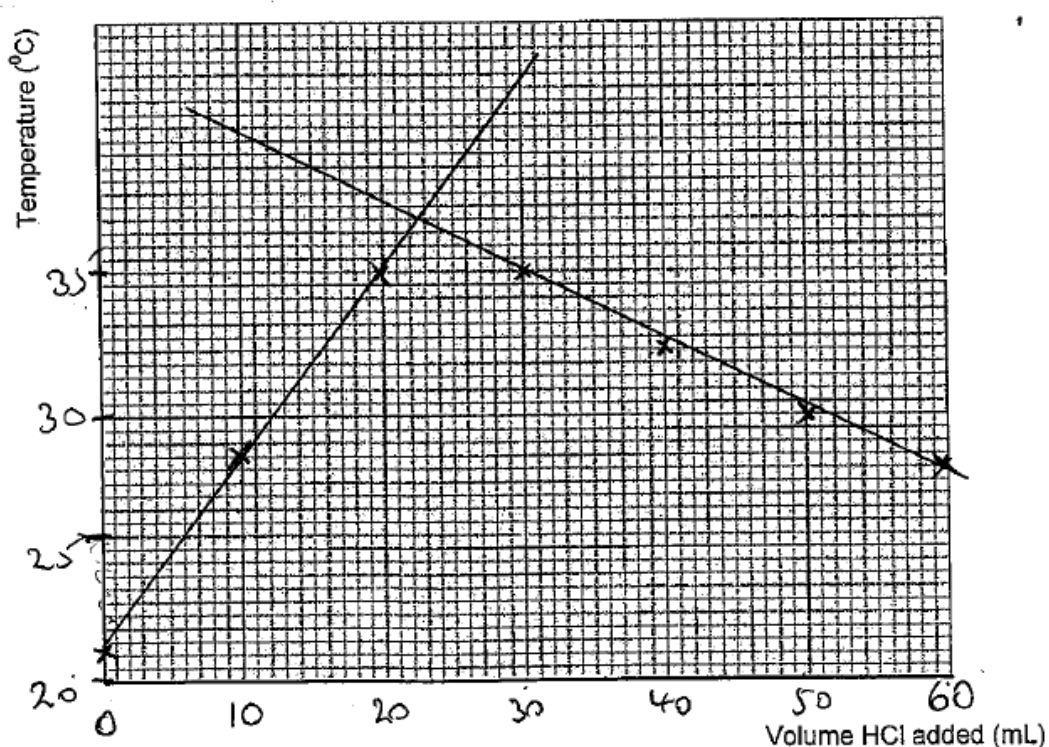
On opening, the pressure is reduced, causing the above equilibrium to move to the left, as concentration of gaseous carbon dioxide decreases. As a result, the hydrogen ion concentration decreases and the pH increases.

Question 29 (10 marks)

29.a.

Marking Guidelines	Marks
Constructs even and appropriately sized scales on axes, and correctly plots all points.	3
Axes or points have one error.	2
Scales or points are correct.	1

Sample answer:



29.b.

Marking Guidelines	Marks
Draws 2 straight lines that cross at about 37.5 mL	1

29.c.

Marking Guidelines	Marks
Identifies the correct volume of HCl required to completely neutralise the NaOH, showing working on the graph.	1

Sample answer:

22 – 23 mL.

29.d.

Marking Guidelines	Marks
Correctly calculates the concentration of the HCl, based on the answer to 25(c).	2
Calculation contains one error.	1

Sample answer:

Volume of HCl is 22.5 mL.

Moles of NaOH used = $C \times V = 0.145 \times 0.025 = 3.625 \times 10^{-3}$

Moles of HCl used therefore = 3.625×10^{-3} (because $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$)

Concentration of HCl = $n/V = 3.625 \times 10^{-3} / 0.0225 = 0.161\text{M}$.

29.e.

Marking Guidelines	Marks
- Calculates grams which match volumetric flask - Describes the method - Correct rinsing - Fills to calibration mark	3
- Calculates grams AND - Describes the method OR - Correct rinsing OR - Fills to calibration mark	2
- Identifies volumetric flask OR - Calculates moles OR - Identifies one part of method	1

Sample answer:

1. A 250 mL volumetric flask was used which was rinsed with distilled water..

2. Calculate the number of grams of $\text{H}_2\text{C}_2\text{O}_4$ needed.

$$n \text{ H}_2\text{C}_2\text{O}_4 = M \times V(\text{L}) = 0.1 \times .25 = .025$$

$$g = n \times \text{MM} = .025 \times (1.002 \times 2 + 12 \times 2 + 16 \times 4) = .025 \times 90.004 = 2.25 \text{ g}$$

3. Weigh out 2.25 g of $\text{H}_2\text{C}_2\text{O}_4$ into a clean beaker.

4. Dissolve in distilled water and decant in to the 250mL volumetric flask.

5. Fill to the calibration mark, stopper and invert.

Question 30 (6 marks)

30.a.

Marking Guidelines	Marks
Identifies a use	1

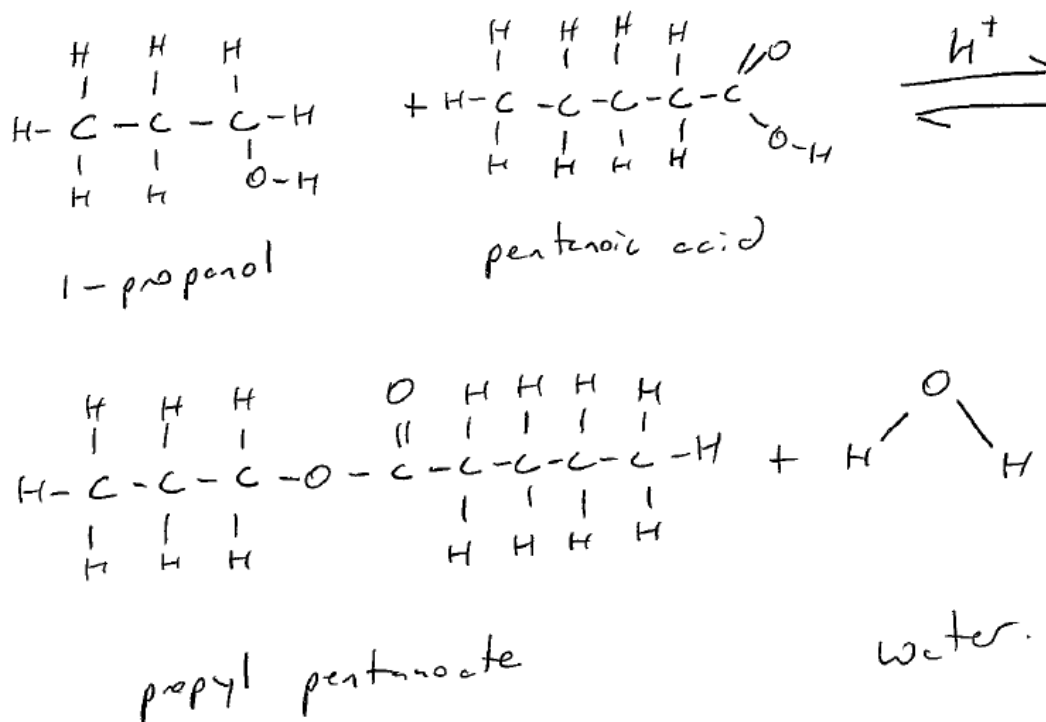
Sample answer:

Flavourings; perfumes; food colourings

30.b.

Marking Guidelines	Marks
Correct equation (must be at equilibrium)	2
- Both reactants correctly drawn OR - Both products correctly drawn	1

Sample answer:



30.c.

Marking Guidelines	Marks
- Identifies refluxing ; catalyst - Amounts of reactants - Draws a labelled diagram	3
- Identifies refluxing or a condenser or catalyst AND - Draws a diagram	2
- Identifies refluxing or a condenser or catalyst OR - Draws a diagram (not labelled)	1

Sample answer:

1. Add 10 mL of 1-propanol and 10 mL of propanoic acid into a flask.
 2. Add 2 drops of concentrated sulphuric acid.
 3. Add 3 boiling chips.
 4. Heat under reflux (see diagram), using a hot water bath, for at least 15 minutes.
- (Note esterification reactions are slow and the reaction usually does not proceed to completion)

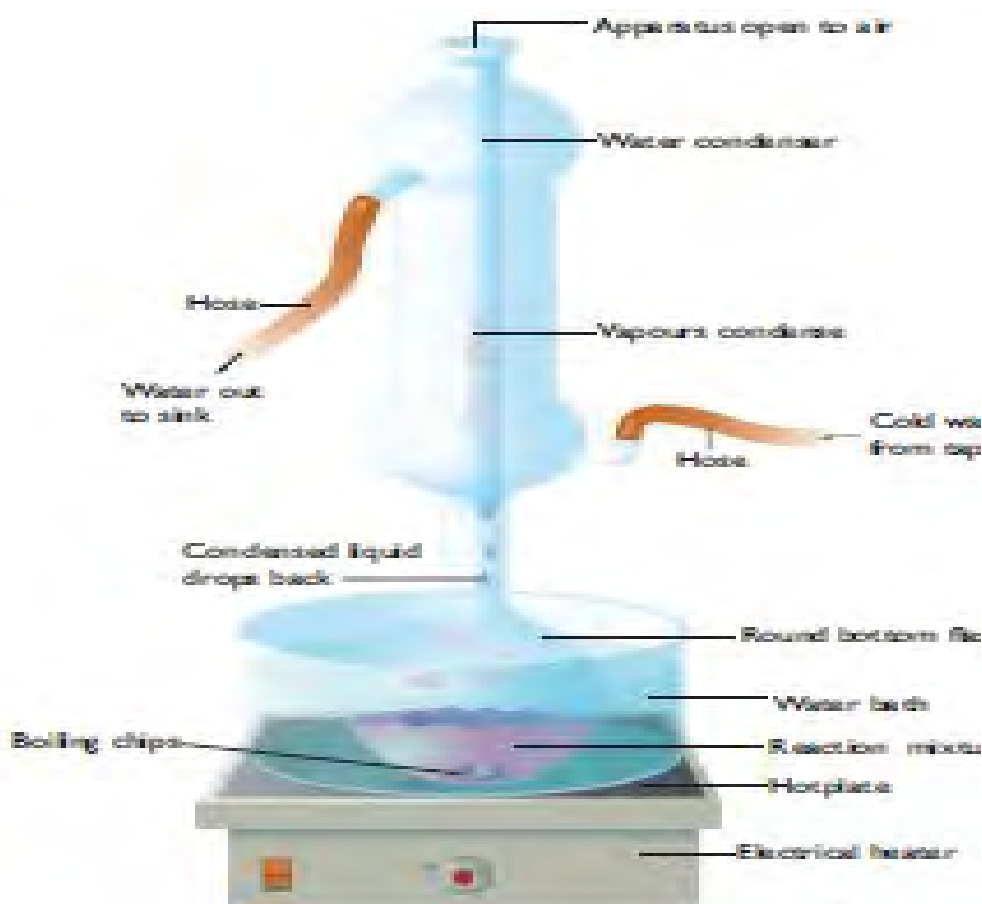


Figure 10.11

The reaction mixture, including a catalyst and boiling chips, is heated under reflux. Vapours condense back to the liquid state and drip back into the reaction vessel.

PART B: HAYES (16 marks)

Question 31 (4 marks)

Marking guidelines

Criteria	Marks
Correctly determines the moles of both reactants AND Correctly identifies that HCl is the limiting reagent (or that there is an excess of Na ₂ S) AND Determines the moles of H ₂ S produced (using the limiting reagent) AND Determines the volume of H ₂ S (based on the moles of H ₂ S calculated) correct to 2 significant figures	4
Correctly determines the moles of both reactants AND Correctly identifies that HCl is the limiting reagent (or that there is an excess of Na ₂ S) AND Determines the moles of H ₂ S produced (using the limiting reagent) AND Determines the volume of H ₂ S (but not correct to 2 significant figures)	3
Any two of the above steps processed correctly	2
Any one of the above steps processed correctly	1

Sample answer

- Moles HCl = $0.025 \times 0.10 = 2.5 \times 10^{-3}$
Moles Na₂S = $0.15 / 2 \times 22.99 + 32.07 = 1.92 \times 10^{-3}$
- $2\text{HCl}_{(\text{aq})} + \text{Na}_2\text{S}_{(\text{s})} \longrightarrow \text{H}_2\text{S}_{(\text{g})} + 2\text{NaCl}_{(\text{aq})}$
HCl is limiting reagent: since 2.5×10^{-3} moles react with 1.25×10^{-3} moles of Na₂S (there is an excess of Na₂S)
- Therefore from equation: Moles of H₂S gas produced = 1.25×10^{-3}
- Volume at 25°C and 100kPa = $1.25 \times 10^{-3} \times 24.79\text{L} = 0.031\text{L}$ or 31mL

(2significant figures)

Outcomes assessed: H2, H6, H13

MARKING GUIDELINES

Criteria	Marks
• Identifies the bonding using a Lewis electron-dot diagram	2
• Draws Lewis electron-dot diagram OR • Identifies the position of the coordinate covalent bond OR • Identifies the position of the double covalent bond	1

Question 32 is now worth 1 mark and so

Criteria	Marks
Identifies the bonding using a correctly drawn Lewis electron dot diagram	1
A correctly drawn electron dot diagram	1/2

Question 33

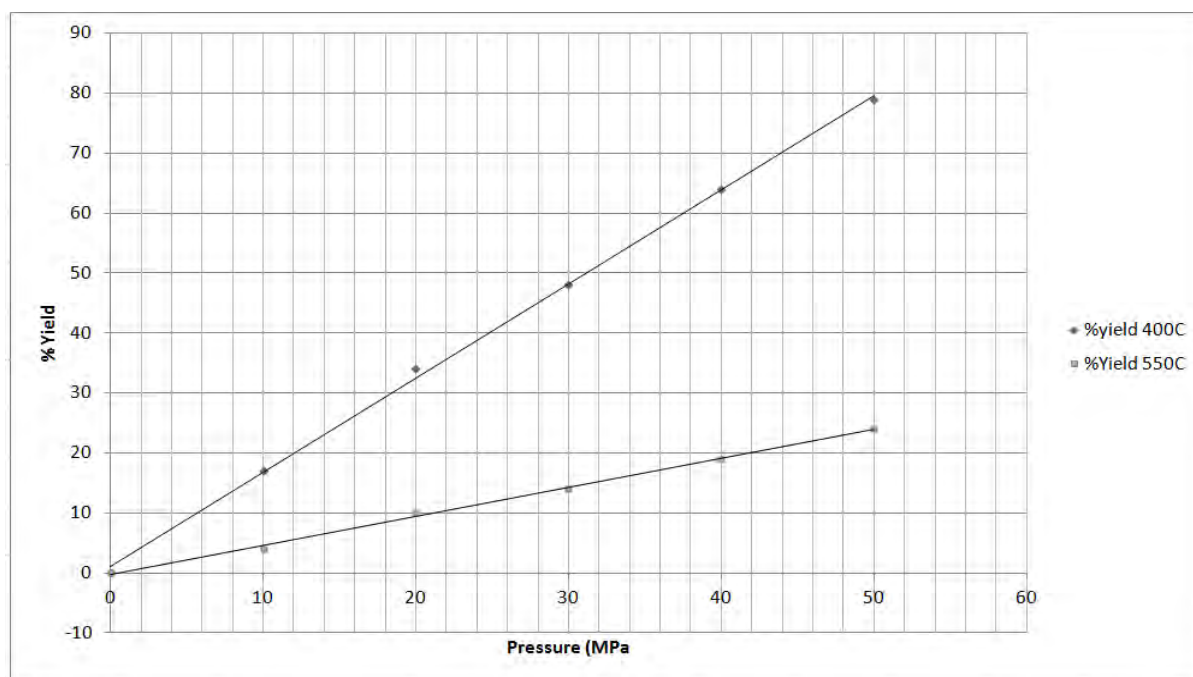
MARKING GUIDELINES

Criteria Marks

Marking criteria for graph

Criteria	Marks
Assigns variables to the correct axis and uses a suitable scale Plots all data points with accuracy Draws and labels both lines of best fit on the same graph	3
Two of the above	2
One of the above	1

Sample graph



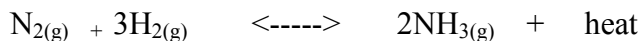
33 (b)

Criteria	Marks
Constructs reaction equation, including an energy term or ΔH as -ve (1 mark)	3
Explains the effect of Temperature (1 mark)	
Explains the effect of Pressure (1 mark)	

Criteria	Marks
Constructs reaction equation, including an energy term (1 mark)	2
Identifies the effect of Temperature (1/2 mark)	
Identifies the effect of pressure (1/2 mark)	

Criteria	Marks
Constructs reaction equation, including an energy term	1
OR identifies the effect of Temperature and pressure	

*1/2 mark awarded here for equation without energy term

Sample answer

Since the forward reaction is exothermic an increase in temperature will cause the equilibrium to shift to the left and therefore decrease the yield of ammonia. Increasing the pressure causes the equilibrium to shift to the right as there are fewer gaseous molecules on the product side and therefore the yield of ammonia is increased.

Question 34**MARKING GUIDELINES****(i)**

Criteria	Marks
Correctly determines concentration of iron from graph	1

(ii)

Criteria	Marks
Correctly calculates the mass of iron	2
Shows one correct step in the calculation	1

Criteria	Marks
Outlines how a calibration curve is obtained	2
Identifies one aspect of the preparation or use of a calibration curve	

Sample answer

(i) 4.5ppm

(ii) answer: 4.5ppm is equal to 4.5mg in 1L

therefore in 600mL there is $0.6 \times 4.5\text{mg} = 2.7\text{mg}$

Sample answer

(iii) To obtain a calibration curve the chemist would need to make a series of standard solutions containing Fe^{2+} , and the range of concentrations should include the expected concentration of the Fe^{2+} in the waterway. Then the absorbance of each solution is measured at the same wavelength and the graph plotted.

PART C: CARRINGTON (15 marks)
OPTION-INDUSTRIAL CHEMISTRY

Question 35 (15 marks)

(a)

(i)

<u>Marking Criteria</u>	<u>Marks</u>
Identifies temperature	1

Sample Answer

Temperature

(ii)

<u>Marking Criteria</u>	<u>Marks</u>
Calculates the numerical value for K with correct working	2
- Identifies the correct K expression OR - Substitutes into an incorrect K expression.	1

Sample answer

$$K = \frac{[CO_2] [H_2]}{[CO] [H_2O]}$$

	<i>CO</i>	<i>H₂O</i>	<i>CO₂</i>	<i>H₂</i>
<i>Initially</i>	1.3	2.4	0	0
<i>At eqm.</i>	0.7	1.8	0.6	0.6

$$K = \frac{[0.6] [0.6]}{[0.7] [1.8]}$$

$$K = 0.2857$$

(b)

<u>Marking Criteria</u>	<u>Marks</u>
Outlines an environmental issue associated with the extraction of sulfur.	1

Sample Answer

Sulfur is present naturally as a solid. When it is extracted from the ground, empty space remains where the sulfur once was. This can lead to surrounding land collapsing into the space, called land subsidence.

OR

The process of mining requires large amounts of land to be cleared, resulting in habitat loss, and ecosystem damage. In extreme cases it could result in a reduction of species diversity.

(c)

Marking Criteria	Marks
- Explains ionisation of sulphuric acid in 2 steps - Includes a balanced equation for one ionisation	2
- Describes ionisation of sulphuric acid OR - Includes a correctly balanced equation for one ionisation	1

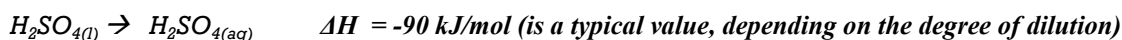
Sample Answer

The ionisation of sulfuric acid is exothermic, releasing lots of heat.

Sulfuric acid ionises in two steps.

Sulfuric acid is a strong acid in its first ionisation, but the HSO_4^- ion formed in the first ionisation is a weak acid and only ionises slightly in the second ionisation.

In concentrated sulphuric acid there are only a few ions. When the acid is added to water it ionises liberating large amounts of heat.



1st Ionisation



K is very large.

2nd Ionisation



In the first step, an H-O bond is broken (endothermic) and a coordinate covalent bond between the hydrogen and a water molecule is formed (exothermic). The exothermic term is much greater than the endothermic term, hence the overall enthalpy change is negative.

The same process occurs in the second ionisation step, but because the H^+ ions has to leave the hydrogen sulfate ion, which already has a negative charge, it is a weak acid and so the reaction occurs to a lesser extent.

(d)

(i)

Marking Criteria	Marks
- Outlines a first -hand investigation to model an equilibrium reaction with equipment and how it demonstrates equilibrium OR - Draws a diagram to explain how it demonstrates the equilibrium model	2
Outlines a first -hand investigation to model an equilibrium reaction OR	1

- | | |
|--|--|
| <ul style="list-style-type: none"> • Draws a diagram to model equilibrium | |
|--|--|

Sample Answer (i)

Two identical measuring cylinders are filled with different volumes of water. Water is transferred backwards and forwards from each cylinder using two differently sized pipettes until the volume of water in each cylinder remains constant (but at different levels).

- There are many models that could 'physically' model equilibrium in a closed system (that are 'non-chemical')
- Diagrams can be included

(ii)

<u>Marking Criteria</u>	<u>Marks</u>
• Describes the requirements for the validity of the experiment	2
• Identifies a requirement for the validity of the experiment	1

Sample answer:

The information collected while physically modelling equilibrium is valid to a point, in that it reflects many of the characteristics of a closed chemical system at equilibrium. These characteristics include initial macroscopic changes in concentrations (volumes of water) of both 'reactants', and 'products' that continue during the transfer of water backwards and forwards until there is no macroscopic change in water levels. This is despite the dynamic nature of the continued movement of water from one cylinder to another. However, the limitations of any model in reflecting all characteristics of an actual chemical equilibrium system does compromise its validity.

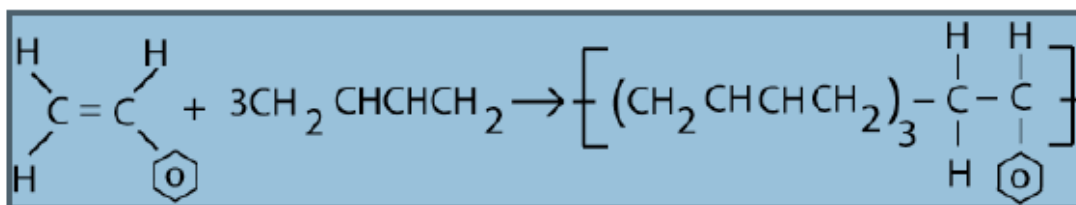
(e)

MARKING GUIDELINES Criteria	Marks
- Identifies an appropriate <u>natural product</u> - Provides characteristics and features of at least <i>TWO</i> issues associated with shrinking world supplies of the natural product AND - Provides characteristics and features of progress being made to find new replacement materials OR <u>Identifies the replacement product</u> - Provides a response that demonstrates coherence and logical progression	5
- Identifies an appropriate <u>natural product</u> AND - Provides characteristics and features of <i>TWO</i> issues associated with shrinking world supplies of the natural product OR - Identifies and provides characteristics and features of one issue associated with shrinking world supplies of the natural product AND - Provides characteristics and features of progress being made to find new replacement materials OR <u>Identifies an appropriate replacement product</u> - Identifies an appropriate <u>natural product</u> AND - Provides characteristics and features of one issue associated with shrinking world supplies of the natural product OR - Provides characteristics and features of progress being made to find new replacement materials OR <u>Identifies an appropriate replacement product</u>	4
- Identifies an appropriate natural product and a replacement material OR - Identifies an appropriate <u>natural product</u> AND - Identifies an issue (renewable/non-renewable) OR <u>Identifies an appropriate replacement product</u> AND - Identifies an issue	3
Identifies an appropriate natural product OR	2

- Identifies a replacement product OR - Identifies an issue (renewable/non-renewable)	1
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Sample Answer

A natural product is one that is used with little or no modification. Raw rubber is an addition polymer that is also a natural product. Raw rubber is obtained from the sap of the rubber tree and is a very useful natural product. It is used in balls, shoes, tyres and as elastic bands. In the early 20th century demand for rubber outstripped supply. The supply of rubber was limited as rubber trees could only produce a certain amount of rubber each year. In addition, this production was mainly from Asia. As the demand for rubber grew, new alternatives needed to be produced. This situation became crucial during the first and second world wars. Rubber was needed for tyres for military vehicles and the limited supply of rubber greatly influenced the war. In the second world war, the Japanese had control of rubber producing areas which resulted in other nations needing scientists to develop synthetic rubber. Another issue that caused the rising demand for rubber was that the automobile industry was growing and car manufacturers needed rubber. Steps to solve the limited supply of rubber involved the development of synthetic rubbers. In the 1950s SBR (styrene butadiene rubber) was formed.



Synthetic rubber improved the properties of natural rubber as the rubber was vulcanised. Short sulfur chains formed crosslinks between polymer chains. This improved the properties of synthetic rubber, making it more durable, more resistant to chemical attack and stronger.

The progress made to improve and increase the supply of rubber has been very effective as the demand for synthetic rubber as a total percentage of rubber is around 80%.

Synthetic rubber has allowed the production of rubber to meet demand and maintain a low cost. New developments to rubber involve the use of non-petrochemicals as the cost of petroleum products has increased.